

ROS 2 Path Planner Benchmarking Report

Performance Evaluation Results — Pass Threshold >= 85.0/100

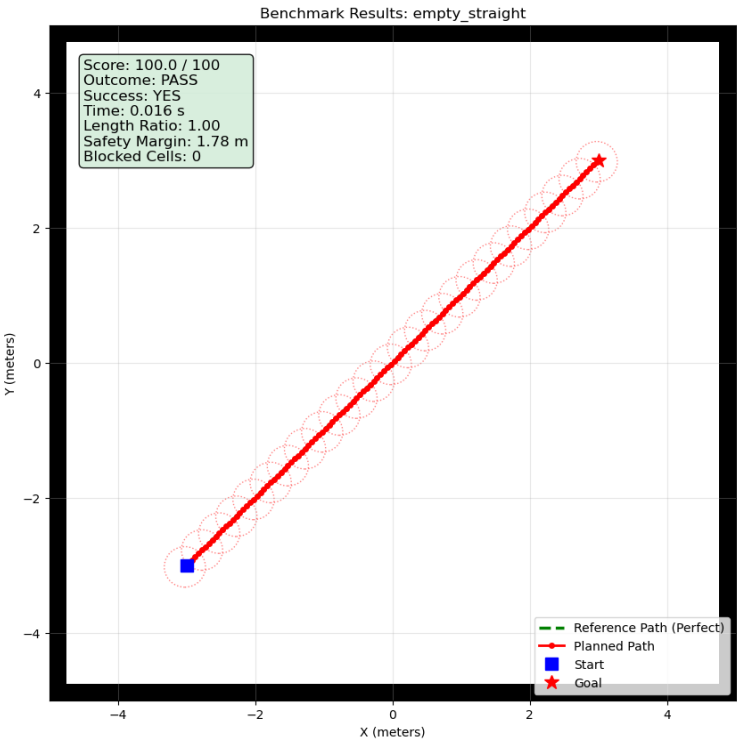
Performance Summary Table

Scenario ID	Outcome	Score	Success	Time	Collisions	Avg Cost	Replan
empty_straight	PASS	100.0/100	YES	0.016 s	0	0.0	100.0
scattered_rocks_detour	FAIL	65.0/100	YES	0.161 s	23	0.0	0.0
canyon_gate_passage	PASS	100.0/100	YES	0.038 s	0	0.0	100.0
marsyard_rough_slopes	PASS	100.0/100	YES	0.523 s	0	0.0	100.0
marsyard_labyrinth	FAIL	65.0/100	YES	0.471 s	241	0.0	0.0
canyon_gate_blocked	PASS	90.0/100	YES	0.019 s	0	0.0	0.0
marsyard_snake_passage	FAIL	65.0/100	YES	0.266 s	32	0.0	0.0
dead_end_trap	FAIL	64.1/100	YES	0.010 s	6	5.9	0.0
crater_field	PASS	100.0/100	YES	0.284 s	0	0.0	100.0

Detailed Scenario Plots & Insights

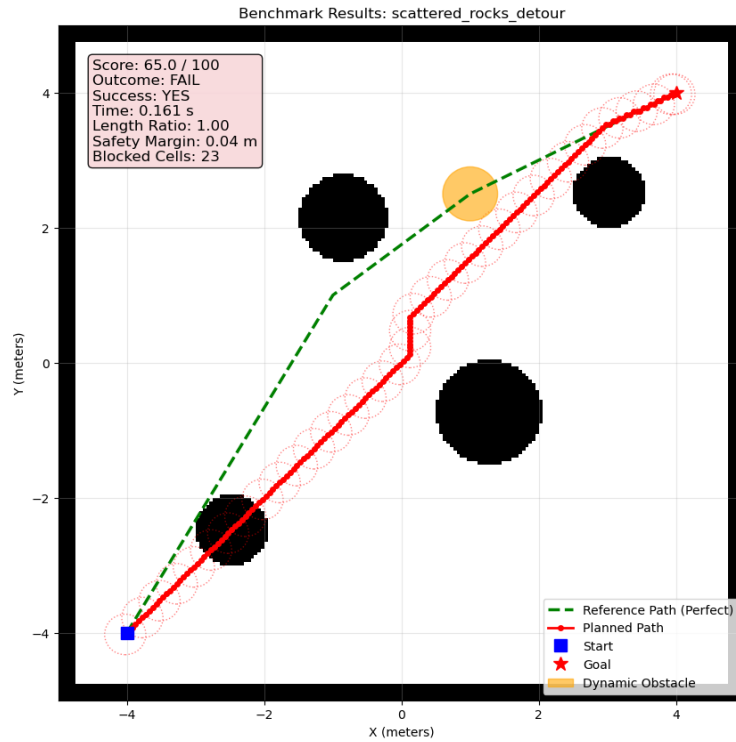
Scenario: empty_straight (PASS)

- Final Score: 100.0/100 | Planning Time: 0.016 s
- Path Length: 8.49 m (Ratio: 1.00) | Safety Margin: 1.78 m
- Footprint Collisions: 0 | Avg Traversed Cost: 0.0
- Replanning Score: 100.0/100 (Time: 0.000 s)



Scenario: scattered_rocks_detour (**FAIL**)

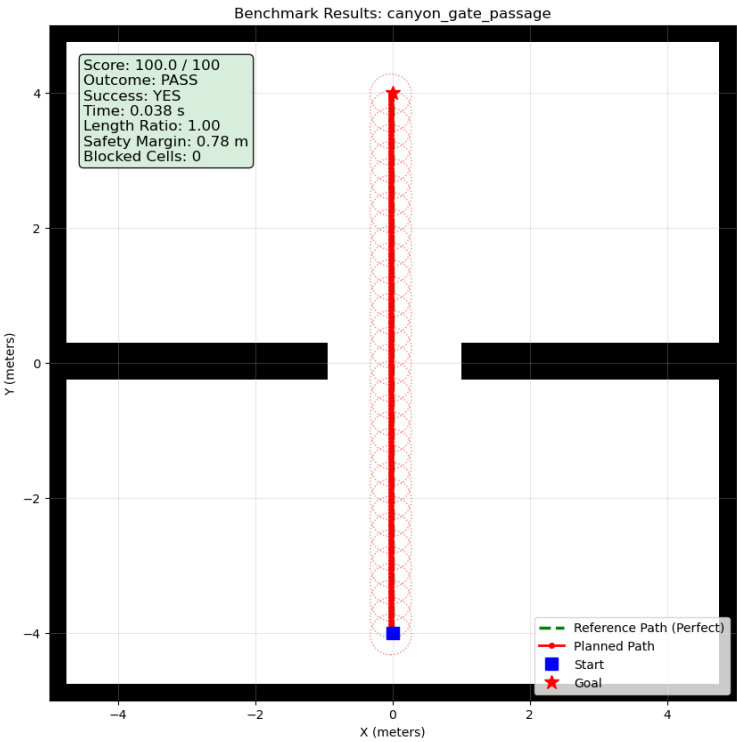
- Final Score: **65.0/100** | Planning Time: 0.161 s
- Path Length: 11.64 m (Ratio: 1.00) | Safety Margin: 0.04 m
- Footprint Collisions: 23 | Avg Traversed Cost: 0.0
- Replanning Score: 0.0/100 (Time: 0.156 s)



Collision Warning: Footprint radius (0.3m) intersected obstacles. Implement costmap obstacle inflation.

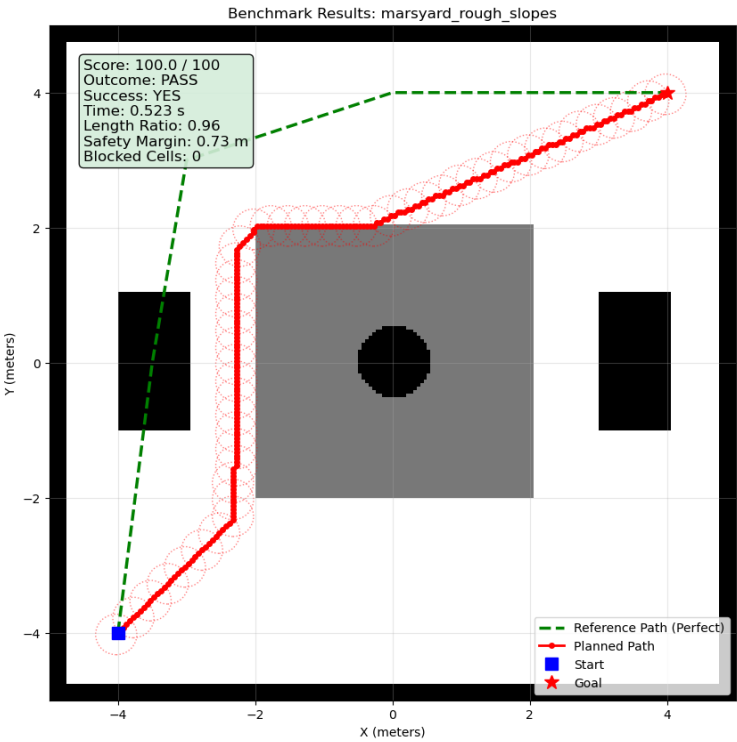
Scenario: canyon_gate_passage (PASS)

- Final Score: **100.0/100** | Planning Time: 0.038 s
- Path Length: 8.00 m (Ratio: 1.00) | Safety Margin: 0.78 m
- Footprint Collisions: 0 | Avg Traversed Cost: 0.0
- Replanning Score: 100.0/100 (Time: 0.000 s)



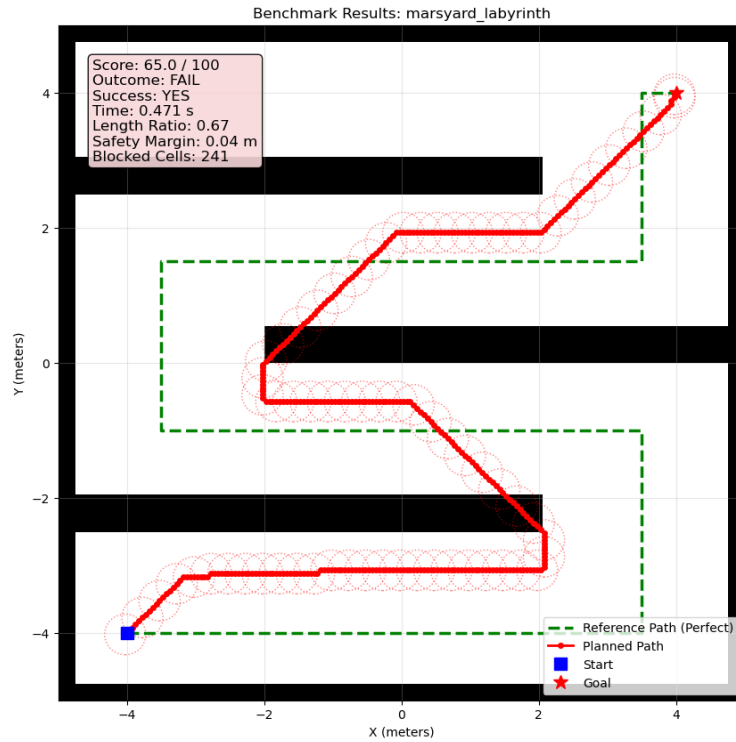
Scenario: marsyard_rough_slopes (PASS)

- Final Score: 100.0/100 | Planning Time: 0.523 s
- Path Length: 13.66 m (Ratio: 0.96) | Safety Margin: 0.73 m
- Footprint Collisions: 0 | Avg Traversed Cost: 0.0
- Replanning Score: 100.0/100 (Time: 0.000 s)



Scenario: marsyard_labyrinth (**FAIL**)

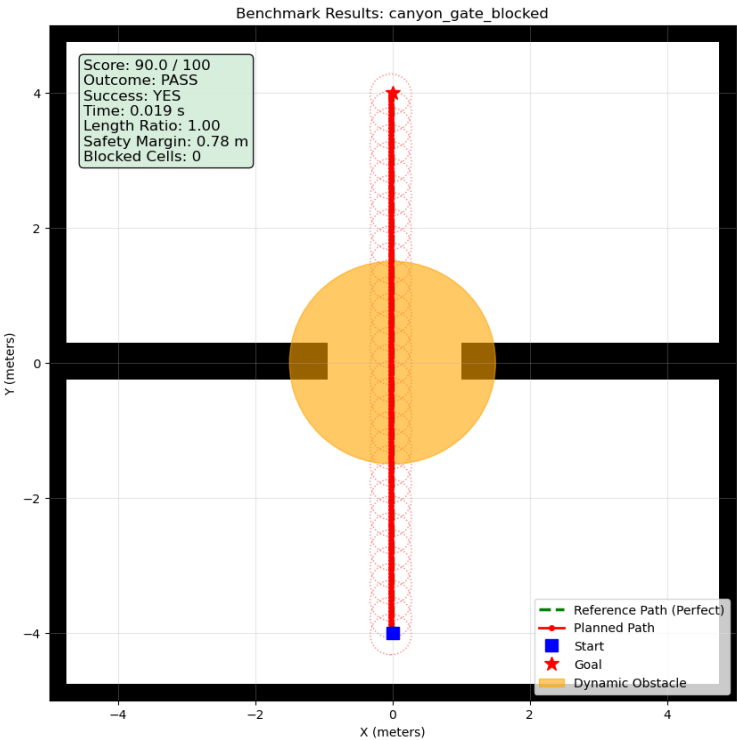
- Final Score: **65.0/100** | Planning Time: 0.471 s
- Path Length: 20.16 m (Ratio: 0.67) | Safety Margin: 0.04 m
- Footprint Collisions: 241 | Avg Traversed Cost: 0.0
- Replanning Score: 0.0/100 (Time: 0.000 s)



Collision Warning: Footprint radius (0.3m) intersected obstacles. Implement costmap obstacle inflation.

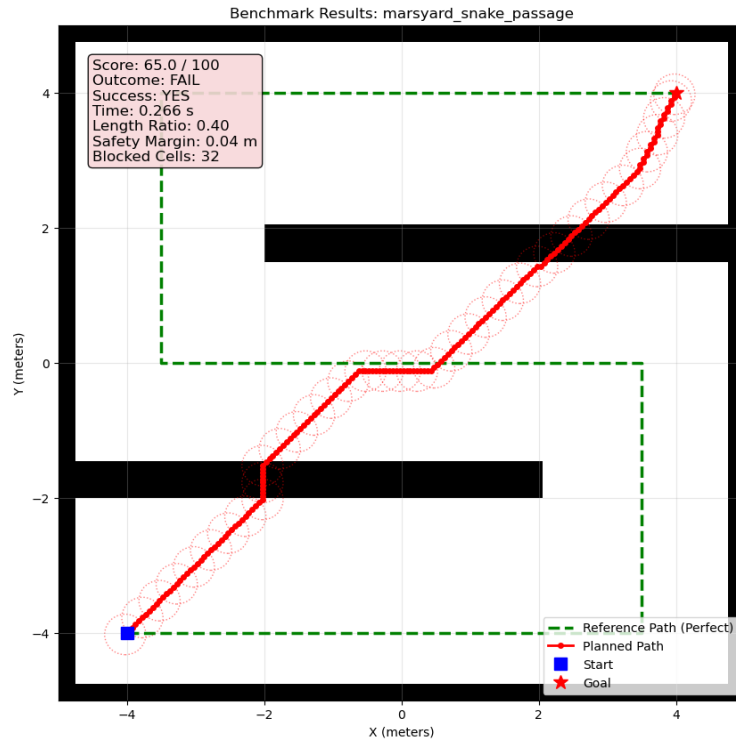
Scenario: canyon_gate_blocked (PASS)

- Final Score: **90.0/100** | Planning Time: 0.019 s
- Path Length: 8.00 m (Ratio: 1.00) | Safety Margin: 0.78 m
- Footprint Collisions: 0 | Avg Traversed Cost: 0.0
- Replanning Score: 0.0/100 (Time: 0.458 s)



Scenario: marsyard_snake_passage (**FAIL**)

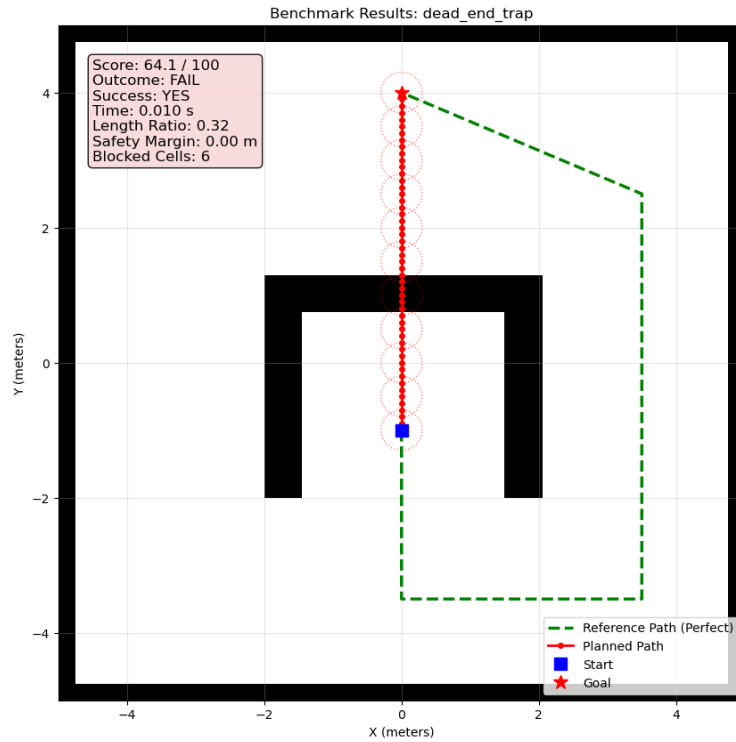
- Final Score: **65.0/100** | Planning Time: 0.266 s
- Path Length: 11.96 m (Ratio: 0.40) | Safety Margin: 0.04 m
- Footprint Collisions: 32 | Avg Traversed Cost: 0.0
- Replanning Score: 0.0/100 (Time: 0.000 s)



Collision Warning: Footprint radius (0.3m) intersected obstacles. Implement costmap obstacle inflation.

Scenario: dead_end_trap (**FAIL**)

- Final Score: **64.1/100** | Planning Time: 0.010 s
- Path Length: 5.00 m (Ratio: 0.32) | Safety Margin: 0.00 m
- Footprint Collisions: 6 | Avg Traversed Cost: 5.9
- Replanning Score: 0.0/100 (Time: 0.000 s)



Collision Warning: Footprint radius (0.3m) intersected obstacles. Implement costmap obstacle inflation.

Scenario: crater_field (PASS)

- Final Score: **100.0/100** | Planning Time: 0.284 s
- Path Length: 12.19 m (Ratio: 0.83) | Safety Margin: 0.78 m
- Footprint Collisions: 0 | Avg Traversed Cost: 0.0
- Replanning Score: 100.0/100 (Time: 0.000 s)

